

Matrix Theory and Computation

Detailed lecture notes

Three sessions, each with two 45-minute periods

First-year postgraduate course

Sections 2.1–2.3 of Golub and Van Loan, *Matrix Computations*

Part 1 Linear systems, subspaces, and low-rank updates

Part 2 Vector norms, error, and spectral geometry

Part 3 Matrix norms, iterative solves, and perturbations

Worked calculations, proof explanations, teaching prompts, and solutions.

Optional study is separated from the scheduled classroom route.

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How to use these notes

These notes accompany three independent Beamer decks for first-year postgraduate students. Each part contains **two 45-minute teaching periods**. The scheduled break is additional to the 90 teaching minutes. Worked calculations, proof explanations, questions, and answer keys support both teaching and independent study.

Prerequisites are matrix multiplication, elementary elimination, determinants, and scalar differentiation. No numerical analysis background is assumed. The classroom activities can be completed by hand; programming appears in an optional workshop.

Part 2 places the eigenvalue material from Section 2.1 after vector norms, preparing students for the spectral norm in Part 3. Example numbers are retained from the earlier 112-slide deck.

Part	Main topic	Scheduled slides	Optional pages	Teaching time
1	Linear systems, subspaces, low-rank updates	1–29	30–37	45 + 45 minutes
2	Vector norms, errors, spectral geometry	1–33	34–39	45 + 45 minutes
3	Matrix norms, iteration, perturbations	1–31	32–49	45 + 45 minutes

Scheduled slides include the title, plan, questions, and break reminder. The first optional page is a divider; the remaining appendix pages are for homework or tutorials, outside the 90-minute lesson.

Teaching pattern. Introduce the question, work one example slowly, and let students try the next calculation before showing the answer. Write dimensions beside unfamiliar matrices, check numerical results in the original equation, and state a theorem's assumptions before using it.

Pacing if students need more time. Keep the independent checks and the final discussion. In Part 1, assign the rank-two arithmetic on slide 26 for homework after explaining the small-system algorithm. In Part 2, use the Rayleigh plot for review after students understand the weighted-average proof. In Part 3, supply the exact spectral eigenvector on slide 15 without asking students to reproduce its decimal normalization. Do not add the optional pages to an already full session.

Notation and conventions

A real matrix $A \in \mathbb{R}^{m \times n}$ takes an input $x \in \mathbb{R}^n$ to an output $Ax \in \mathbb{R}^m$. The notation $a_j = A(:, j)$ denotes column j . A coordinate vector e_j has a one in position j and zeros elsewhere. Transpose is written A^T . For complex matrices, the corresponding inner-product formulas use conjugate transpose A^* ; the worked examples here are real unless explicitly stated otherwise.

The range and null space are

$$\text{ran}(A) = \{Ax : x \in \mathbb{R}^n\}, \quad \text{null}(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

A statement involving A^{-1} assumes that A is square and nonsingular. In an algorithm, the notation $A^{-1}b$ usually means **solve** $Ay = b$. It does not instruct us to compute every entry of the inverse.

We use the textbook convention that a norm on the matrix vector space need not automatically satisfy a product inequality. Induced matrix norms do satisfy the appropriate product bound. Some other texts include that bound in their definition of a matrix norm; check the convention before comparing statements.

Part 1. Linear systems, subspaces, and low-rank updates

Learning outcomes and teaching plan

Students should be able to find a column-space basis from elimination, describe all solutions of a consistent system, compute an orthonormal basis and a projection, and solve a low-rank updated system by reusing solves with its original matrix. They should also identify the assumptions needed by the update formula.

Period 1: structure of a linear system.

Minutes	Slides	Teaching and student work
0–4	1–2	State the aim; recall the input/output dimensions of a rectangular matrix.
4–10	3–5	Independence, span, and a redundant set of columns.
10–23	6–10	Elimination, original pivot columns, all solutions, rank factorization.
23–35	11–14	Orthogonality, Gram–Schmidt, and projection.
35–41	15–16	Four subspaces and the consistency condition.
41–45	17	Individual solvability check and feedback.

Slide 18 marks the break. **Period 2: reuse after a matrix update.**

Minutes	Slides	Teaching and student work
0–5	19–20	Inverse actions and reversal of multiplication order.
5–14	21–22	Derive the Sherman–Morrison solve from a scalar unknown.
14–24	23–24	Complete the numerical example and discuss a small denominator.
24–34	25–26	Derive the Woodbury small system and calculate its rank-two example.
34–42	27–28	Independent rank-one solve, verification, and feedback.
42–45	29	Exit questions and preparation for vector norms.

Opening explanation

Suggested opening. “A matrix describes which outputs can be produced from a collection of inputs. Some different inputs may give the same output. We will identify this freedom and use it to solve systems. Later we will ask what happens when the matrix changes in only a few directions.”

Ask what dimensions are required in $Ax = b$ when A is 3×2 . The answer is $x \in \mathbb{R}^2$ and $b \in \mathbb{R}^3$. This quick check prevents a common confusion: the null space belongs to the input space, while the range belongs to the output space.

Independence, span, and a basis

Part 1 slides 3–5.

A linear combination of vectors $a_1, \dots, a_n$ is $\sum_j c_j a_j$. Their span is the set of all such combinations. Independence means that the only coefficients producing zero are all zero. A basis has two separate properties: it is independent, and it spans the space under discussion.

If one generating vector is a combination of the others, removing it leaves the span unchanged. Conversely, an independent list can be enlarged until it spans the target subspace. The number of vectors in a basis is the dimension. It is not necessarily the number of vectors in the original list.

Example 1: three columns spanning a plane

Use the same matrix throughout the first period:

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

First let students inspect the entries. Since $a_3 = a_1 + a_2$, the coefficient vector $(-1, -1, 1)^T$ produces zero. The three columns are dependent.

Next test whether the first two columns are independent. If $\alpha a_1 + \beta a_2 = 0$, the third equation is $\beta = 0$. The first equation then gives $\alpha = 0$. Thus these two columns form a basis for the span.

Their general combination is

$$\alpha a_1 + \beta a_2 = \begin{bmatrix} \alpha + 2\beta \\ 2\alpha + 4\beta \\ \beta \end{bmatrix} = \begin{bmatrix} s \\ 2s \\ t \end{bmatrix}, \quad s = \alpha + 2\beta, \quad t = \beta.$$

Every output satisfies the equation $y_2 = 2y_1$. Conversely, for any s, t , choosing $\beta = t$ and $\alpha = s - 2t$ produces $(s, 2s, t)^T$. We have therefore proved both inclusions in the description

$$\text{ran}(A) = \{(s, 2s, t)^T : s, t \in \mathbb{R}\}.$$

Ask: “Is this a line, a plane, or all of $\mathbb{R}^3$?” It is a plane through the origin, with two independent parameters. Three output coordinates do not imply three independent degrees of freedom.

Elimination, rank, and all solutions

Part 1 slides 6–10.

The rank is the dimension of the range. Elimination reveals the number of independent constraints and the positions of pivot columns. For the running example, write each operation explicitly:

$$\begin{aligned} A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix} &\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \\ &\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

There are two pivots, in columns 1 and 2. Hence $\text{rank}(A) = 2$, and the original columns a_1, a_2 form a basis of $\text{ran}(A)$.

Why the original columns? Row operations multiply A on the left by an invertible matrix E . Thus $R = EA$. The relation $Rc = 0$ holds exactly when $Ac = 0$, so dependence relations and pivot positions are preserved. However, output vectors are transformed by E . The column space of R need not equal that of A . Here the first two columns of R span the plane $y_3 = 0$, while the original columns span $y_2 = 2y_1$.

The null space

The reduced homogeneous system gives

$$x_1 + x_3 = 0, \quad x_2 + x_3 = 0.$$

Choose the free variable $x_3 = t$. Then

$$x = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \text{null}(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

The nullity is one. Rank plus nullity is $2 + 1 = 3$, the number of input coordinates.

Interpretation. Adding the vector $t(-1, -1, 1)^T$ to an input leaves the output unchanged. In a linear prediction model, two parameter vectors differing by this null vector make the same predictions on the represented observations.

Solving a nonhomogeneous system

Take $b = (4, 8, 1)^T$. Apply the same row operations to the augmented matrix:

$$[A \mid b] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $x_1 + x_3 = 2$ and $x_2 + x_3 = 1$. The choice $x_3 = 0$ gives a particular solution $x_p = (2, 1, 0)^T$. All solutions are

$$x = x_p + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-t \\ 1-t \\ t \end{bmatrix}.$$

Check directly:

$$Ax = \begin{bmatrix} (2-t) + 2(1-t) + 3t \\ 2(2-t) + 4(1-t) + 6t \\ (1-t) + t \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}.$$

The formula “particular solution plus null space” is exact whenever a particular solution exists. If b is outside the range, there is no particular solution and the formula cannot be used to manufacture one.

Rank factorization

Since $a_3 = a_1 + a_2$,

$$A = XY^T, \quad X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}, \quad Y^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

To multiply Ax , first form $c = Y^T x = (x_1 + x_3, x_2 + x_3)^T$, then form Xc . The calculation passes through a two-dimensional intermediate space. Both factors have rank two. This is an exact representation of this rank-two matrix; it does not yet solve the problem of finding the best low-rank approximation to an arbitrary matrix.

Orthogonality and projections

Part 1 slides 11–14.

Vectors are orthogonal when their inner product is zero. An orthonormal list is orthogonal and each vector has Euclidean length one. Nonzero mutually orthogonal vectors are independent: if $\sum_j c_j q_j = 0$, taking the inner product with q_i gives $c_i \|q_i\|_2^2 = 0$, so $c_i = 0$.

For a square matrix Q with orthonormal columns, $Q^T Q = I$ and $Q^{-1} = Q^T$. A rectangular matrix V with orthonormal columns satisfies $V^T V = I$, but $V V^T$ is generally a projection rather than the identity on the larger output space.

Example 5: constructing an orthonormal basis

Start with $a_1 = (1, 2, 0)^T$ and $a_2 = (2, 4, 1)^T$. Normalize the first:

$$\|a_1\|_2 = \sqrt{1+4} = \sqrt{5}, \quad q_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Compute the component of a_2 along q_1 :

$$q_1^T a_2 = \frac{2+8}{\sqrt{5}} = 2\sqrt{5}, \quad q_1(q_1^T a_2) = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}.$$

Remove that component:

$$w_2 = a_2 - q_1(q_1^T a_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

This vector already has length one, so $q_2 = w_2$. A third vector perpendicular to both is

$$q_3 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}.$$

The inner products are $q_1^T q_2 = 0$, $q_2^T q_3 = 0$, and $q_1^T q_3 = (-2+2)/5 = 0$. Their lengths are all one. Therefore $Q = [q_1 \ q_2 \ q_3]$ satisfies $Q^T Q = I_3$.

Explain the two operations separately. Subtracting the projection changes the direction so that it is perpendicular to the previous basis vector. Dividing by the norm changes its length to one. These are different purposes. This is Gram–Schmidt in exact arithmetic; its floating-point behavior will require a separate discussion later in the course.

Example 5 continued: the closest point on a line

Let $x = (3, 1)^T$ and let the line be spanned by $u = (1, 1)^T$. Every point on the line is tu . Minimize the squared distance:

$$f(t) = \|x - tu\|_2^2 = (3-t)^2 + (1-t)^2 = 2t^2 - 8t + 10.$$

Since $f'(t) = 4t - 8$ and $f''(t) = 4 > 0$, the unique minimizer is $t = 2$. Thus

$$p = 2u = (2, 2)^T, \quad r = x - p = (1, -1)^T.$$

The residual is perpendicular to the line because $u^T r = 1 - 1 = 0$. The squared lengths verify Pythagoras:

$$\|x\|_2^2 = 10, \quad \|p\|_2^2 = 8, \quad \|r\|_2^2 = 2.$$

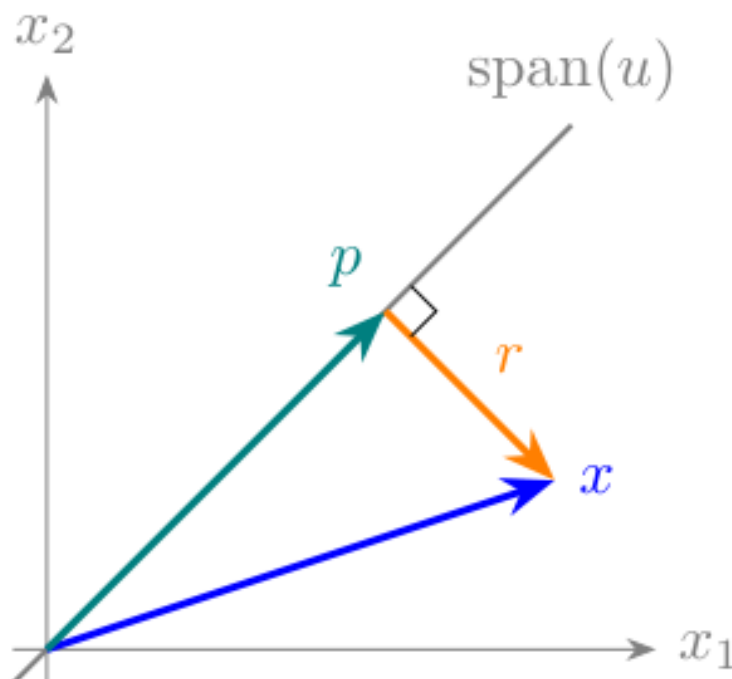


Figure 1: Projection of a vector onto a line. The residual is perpendicular to the line.

For a general nonzero u , expanding the same quadratic gives

$$t_* = \frac{u^T x}{u^T u}, \quad p = u \frac{u^T x}{u^T u}.$$

If $q = u/\|u\|_2$ is a unit vector, this becomes $p = qq^T x$. For an orthonormal basis matrix V of a higher-dimensional subspace, the corresponding formula is $p = VV^T x$.

Ask: “Why is $uu^T x$ generally wrong?” The vector u need not have unit length. The denominator $u^T u$ is essential unless it equals one.

Four fundamental subspaces and solvability

Part 1 slides 15–17.

A vector w is perpendicular to every output of A exactly when

$$w^T Ax = 0 \quad \text{for all } x.$$

This is equivalent to $A^T w = 0$. Therefore

$$\text{ran}(A)^\perp = \text{null}(A^T).$$

The other pair follows by replacing A with A^T .

Space	Ambient space	Dimension for rank r
$\text{ran}(A)$	$\mathbb{R}^m$	r
$\text{null}(A^T)$	$\mathbb{R}^m$	$m - r$
$\text{ran}(A^T)$	$\mathbb{R}^n$	r
$\text{null}(A)$	$\mathbb{R}^n$	$n - r$

For Example 1 the bases are, respectively,

$$\begin{aligned}\text{ran}(A) &: (1, 2, 0)^T, (2, 4, 1)^T, \\ \text{null}(A^T) &: (-2, 1, 0)^T, \\ \text{ran}(A^T) &: (1, 2, 3)^T, (0, 1, 1)^T, \\ \text{null}(A) &: (-1, -1, 1)^T.\end{aligned}$$

Check one inner product in each ambient space. For example, $(-2, 1, 0)^T a_1 = -2 + 2 = 0$, and $(1, 2, 3)(-1, -1, 1)^T = -1 - 2 + 3 = 0$.

The system $Ax = b$ is consistent precisely when $w^T b = 0$ for every $w \in \text{null}(A^T)$. Here that condition is $-2b_1 + b_2 = 0$. For $b = (4, 9, 1)^T$, it fails because $-8 + 9 = 1$. For $b = (4, 8, 1)^T$, it holds, and the solution family has already been calculated.

Period 1 feedback. A correct answer should name the left null vector, show its product with b , and distinguish “no solution” from “many solutions.” A zero determinant alone does not make a square system inconsistent; it means uniqueness fails if a solution exists.

Inverse actions and multiplication order

Part 1 slides 19–20.

For nonsingular square matrices, $(AB)^{-1} = B^{-1}A^{-1}$. The cancellation is

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I.$$

Associativity allows the parentheses to move. It does not permit arbitrary exchanges of factors.

Example 2: an order that can be checked

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad AB = \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}.$$

Then

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}.$$

The correct product is

$$B^{-1}A^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 0 & 1/3 \end{bmatrix}.$$

The upper-right entry of $(AB)(B^{-1}A^{-1})$ is $2(-1/2) + 3(1/3) = 0$. By contrast, $A^{-1}B^{-1}$ has upper-right entry $-1/3$. Multiplying this candidate by AB produces upper-right entry $2(-1/3) + 3(1/3) = 1/3$, so it is not the inverse.

In a large computation, solve with a factorization of A or B instead of forming these inverses. Small explicit inverses are used here only to make the identities visible.

Sherman–Morrison as a solve algorithm

Part 1 slides 21–24.

Suppose we know how to solve systems with a nonsingular A and now need

$$(A + uv^T)x = b.$$

The outer product uv^T has rank at most one. Its action is $uv^T x = u(v^T x)$: one scalar extracted from x controls the correction.

Example 3: derivation before memorization

Define y and z by the solves $Ay = b$ and $Az = u$. Multiplying the updated equation by A^{-1} gives

$$x + z(v^T x) = y.$$

Introduce the scalar $s = v^T x$. Then $x = y - zs$. Multiplying by v^T gives

$$s = v^T y - (v^T z)s, \quad (1 + v^T z)s = v^T y.$$

If $\alpha = 1 + v^T z \neq 0$, then

$$s = \frac{v^T y}{\alpha}, \quad x = y - z \frac{v^T y}{1 + v^T z}.$$

The steps are two solves with A , two dot products, one scalar division, and one vector update. If a dense factorization of A is already available, each new solve has an $O(n^2)$ cost. The savings depend on that reuse; factoring A from scratch is not free.

Complete arithmetic

Take $A = \text{diag}(2, 3)$, $u = v = (1, 1)^T$, and $b = (1, 2)^T$. The updated matrix is

$$A + uv^T = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}.$$

The two diagonal solves give

$$y = \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix}, \quad z = \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix}.$$

Hence

$$v^T z = \frac{5}{6}, \quad \alpha = \frac{11}{6}, \quad v^T y = \frac{7}{6}, \quad s = \frac{7/6}{11/6} = \frac{7}{11}.$$

Now calculate each entry of the update:

$$x_1 = \frac{1}{2} - \frac{1}{2} \frac{7}{11} = \frac{2}{11}, \quad x_2 = \frac{2}{3} - \frac{1}{3} \frac{7}{11} = \frac{5}{11}.$$

Verify using the updated matrix:

$$\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 2/11 \\ 5/11 \end{bmatrix} = \begin{bmatrix} (6+5)/11 \\ (2+20)/11 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The inverse formula follows because the solve formula holds for every b :

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}.$$

For this example,

$$(A + uv^T)^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} - \frac{6}{11} \begin{bmatrix} 1/4 & 1/6 \\ 1/6 & 1/9 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & -1 \\ -1 & 3 \end{bmatrix}.$$

Why a small denominator matters

Take $A = I_2$, $u = e_1$, and $v = -(1 - \varepsilon)e_1$. Then

$$\alpha = \varepsilon, \quad A + uv^T = \text{diag}(\varepsilon, 1).$$

At $\varepsilon = 0$ the updated matrix is singular. For nonzero ε , its inverse is $\text{diag}(1/\varepsilon, 1)$. The large factor is a property of the updated problem, not merely an inconvenient division in the formula. This example explains why checking only the dimensions is insufficient.

Ask: “Does $\alpha \neq 0$ prove a floating-point implementation is accurate?” It proves the algebraic inverse exists under the assumptions. Accuracy also depends on conditioning and rounding in the solves, dot products, and subtraction.

Woodbury: a small system for several update directions

Part 1 slides 25–26.

Example 4: derivation and dimensions

For $U, V \in \mathbb{R}^{n \times k}$, consider $(A + UV^T)x = b$. Solve

$$Ay = b, \quad AW = U.$$

The second equation means k right-hand sides and gives $W \in \mathbb{R}^{n \times k}$. The updated equation becomes $x = y - W(V^T x)$. Define $c = V^T x \in \mathbb{R}^k$. It satisfies

$$\underbrace{(I_k + V^T W)}_{S \in \mathbb{R}^{k \times k}} c = V^T y.$$

Solve this small system, then return $x = y - Wc$. The method requires both A and S to be nonsingular. It replaces a newly changed large system by repeated solves with the original matrix and one small system. When $k \ll n$ and the original factorization is reusable, this can reduce the work substantially.

A rank-two calculation

Let

$$A = 2I_3, \quad U = V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

First, $y = (1/2, 1/2, 0)^T$ and $W = U/2$. Compute

$$U^T U = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad S = I_2 + \frac{1}{2}U^T U = \begin{bmatrix} 2 & 1/2 \\ 1/2 & 2 \end{bmatrix}.$$

The right-hand side is $U^T y = (1/2, 1/2)^T$. The two equations are

$$2c_1 + \frac{1}{2}c_2 = \frac{1}{2}, \quad \frac{1}{2}c_1 + 2c_2 = \frac{1}{2}.$$

Subtracting gives $\frac{3}{2}(c_1 - c_2) = 0$, so $c_1 = c_2$. Substitution gives $\frac{5}{2}c_1 = \frac{1}{2}$, hence $c = (1/5, 1/5)^T$. Therefore

$$Wc = \begin{bmatrix} 1/10 \\ 1/10 \\ 1/5 \end{bmatrix}, \quad x = y - Wc = \begin{bmatrix} 2/5 \\ 2/5 \\ -1/5 \end{bmatrix}.$$

Finally,

$$A + UU^T = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}.$$

Its product with x has entries $6/5 - 1/5 = 1$, $6/5 - 1/5 = 1$, and $2/5 + 2/5 - 4/5 = 0$. This checks the full result without an explicit 3×3 inverse.

Period 2 practice and answers

Part 1 slides 27–29.

For $A = I_2$, $u = v = (1, 1)^T$, and $b = (1, 0)^T$, the two solves give $y = (1, 0)^T$ and $z = (1, 1)^T$. Then $v^T z = 2$, $\alpha = 3$, and $v^T y = 1$. Thus

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -1/3 \end{bmatrix}.$$

The updated matrix is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Its product with x is $(1, 0)^T$. A complete solution must verify this updated system rather than the original identity system.

The exit questions ask students to recall original pivot columns, the null space as the freedom in a consistent system, and the significance of a small update denominator. Ask for reasons or a counterexample, not only the name of a formula.

Optional study for Part 1

Part 1 slides 30–37. This material is outside the scheduled lesson.

Subspaces, affine sets, and direct sums

The plane $y_2 = 2y_1$ is a subspace because it contains zero and is closed under addition and scalar multiplication. The plane $y_2 = 2y_1 + 1$ is not a subspace because it does not contain zero. A direct sum $S_1 \oplus S_2$ requires a unique decomposition $x = x_1 + x_2$, with $x_i \in S_i$. For two subspaces this is equivalent to $S_1 \cap S_2 = \{0\}$: the difference between two decompositions would lie in the intersection.

Why rank–nullity holds

Choose a basis $v_1, \dots, v_k$ of the null space and extend it to a basis $v_1, \dots, v_n$ of the input space. Every output is a combination of $Av_{k+1}, \dots, Av_n$ because the first k basis vectors map to zero. These remaining images are independent. If a combination mapped to zero, the corresponding combination of $v_{k+1}, \dots, v_n$ would belong to the span of $v_1, \dots, v_k$, contradicting independence of the full basis. Thus the rank is $n - k$.

Example 6: determinant size and sensitivity

For $A = 10^{-8}I_2$, $\det(A) = 10^{-16}$, but both directions scale identically. Its 2-norm condition number is $\|A\|_2\|A^{-1}\|_2 = 1$. For $B = \text{diag}(10^8, 10^{-8})$, the determinant is one, but the corresponding condition number is 10^{16} . Determinant zero detects exact singularity. Determinant magnitude alone does not measure relative sensitivity.

Condition numbers are introduced here as a preview and are used more carefully in Part 3. A large condition number describes possible worst-case amplification. It does not say that every right-hand side or every perturbation suffers that amplification.

Example 7: differentiating an inverse

Let $A(t)$ be differentiable and nonsingular near the parameter value. Differentiate $A(t)A(t)^{-1} = I$:

$$A'A^{-1} + A(A^{-1})' = 0.$$

Left multiplication by A^{-1} yields

$$(A^{-1})' = -A^{-1}A'A^{-1}.$$

For $A(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix},$

$$A^{-1} = \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix}, \quad A' = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Multiplication gives $-A^{-1}A'A^{-1} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$, agreeing with direct differentiation. Keep the factor order: in general A' does not commute with A^{-1} .

For $\phi(x) = \frac{1}{2}x^T Ax - b^T x$, expand $\phi(x + th)$ and collect the coefficient of t :

$$\phi(x + th) - \phi(x) = t h^T \left[\frac{1}{2}(A + A^T)x - b \right] + O(t^2).$$

Therefore $\nabla\phi(x) = \frac{1}{2}(A + A^T)x - b$. For $A = \begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix}$, $b = (1, 2)^T$, and $x = (1, 1)^T$, the gradient is $(5/2, 7/2)^T$. The formula $Ax - b$ requires symmetry. A stationary point also requires suitable curvature conditions to be a minimum.

Additional rank exercise

For $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, the second row is twice the first. Hence rank is one and the nullity is $3 - 1 = 2$.

Solving $x_1 + 2x_2 + 3x_3 = 0$ gives

$$\text{null}(B) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The two free variables account for two independent null vectors. The number of rows is not the dimension used on the right side of rank-nullity.

Part 2. Vector norms, error, and spectral geometry

Learning outcomes and teaching plan

Students should be able to calculate and normalize vectors in different norms, prove the basic norm comparisons, distinguish global and componentwise errors, and derive the Rayleigh quotient bounds for a real symmetric matrix. They should be able to explain why a theorem about fixed dimension does not automatically control a sequence of finer grids or larger data sets.

Period 1: measuring vectors.

Minutes	Slides	Teaching and student work
0–4	1–2	Recall the projection example and introduce quantitative error measures.
4–13	3–6	Norm axioms, three common norms, normalization, counterexamples.
13–18	7	Unit-ball geometry and the direction of inclusions.
18–30	8–10	Hölder, a dot-product example, and the Cauchy–Schwarz proof.
30–40	11–14	Norm equivalence, sharp dimension factors, MAE and RMSE.
40–45	15	Independent error-scaling calculation and feedback.

Slide 16 marks the break. **Period 2: error and symmetric geometry.**

Minutes	Slides	Teaching and student work
0–8	17–19	Absolute, relative, and componentwise error calculations.
8–14	20–22	Fixed versus changing dimension; orthogonal invariance.
14–19	23–24	Eigenvectors, eigenvalues, and change of coordinates.
19–31	25–28	Symmetric diagonalization and the Rayleigh weighted-average proof.
31–37	29–30	Interpret the Rayleigh curve and calculate a mixed direction.
37–42	31–32	Practice questions and answers.
42–45	33	Exit questions and preparation for the matrix 2-norm.

Norms as measurements

Part 2 slides 3–7.

Suggested explanation. “The expression ‘small error’ has no numerical meaning until we choose a measurement. A vector error may have one large component or many small components. Different norms make these patterns visible in different ways.”

A vector norm obeys positivity, absolute homogeneity, and the triangle inequality:

$$\|x\| \geq 0, \quad \|x\| = 0 \iff x = 0, \quad \|\alpha x\| = |\alpha| \|x\|, \quad \|x + y\| \leq \|x\| + \|y\|.$$

The distance between two vectors is $d(x, y) = \|x - y\|$. Positivity ensures that distinct vectors have positive distance. Homogeneity controls changes of scale. The triangle inequality bounds the accumulated size of errors that add.

For $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

The most frequently used cases are the sum of absolute entries, Euclidean length, and maximum absolute entry:

$$\|x\|_1 = \sum_i |x_i|, \quad \|x\|_2 = \sqrt{\sum_i x_i^2}, \quad \|x\|_\infty = \max_i |x_i|.$$

Example 10: normalization depends on the norm

For $x = (3, -4, 0)^T$,

$$\|x\|_1 = 3 + 4 = 7, \quad \|x\|_2 = \sqrt{9 + 16} = 5, \quad \|x\|_\infty = 4.$$

The sign of the second entry does not cancel the first entry in the 1-norm. Let $q = x/5 = (3/5, -4/5, 0)^T$. Then $\|q\|_2 = 1$, but $\|q\|_1 = 7/5$ and $\|q\|_\infty = 4/5$. Calling a vector “unit” requires a specified norm.

To normalize in the 1-norm use $x/7$; to normalize in the infinity-norm use $x/4$. The zero vector cannot be normalized by division because its norm is zero.

Example 11: useful penalties that are not norms

The squared Euclidean length $f(x) = \|x\|_2^2$ is nonnegative and widely used in least-squares objectives, but it is not a norm. For $x = e_1$,

$$f(2x) = 4, \quad |2|f(x) = 2.$$

Homogeneity fails. It also fails the triangle inequality for $x = y = e_1$, since $f(x + y) = 4 > 2 = f(x) + f(y)$.

For the expression with $p = 1/2$,

$$g(x) = \left(|x_1|^{1/2} + |x_2|^{1/2}\right)^2,$$

take $x = e_1$ and $y = e_2$. Then $g(x) = g(y) = 1$, while $g(x + y) = 4$. This explicitly shows why the displayed p -norm family assumes $p \geq 1$. A penalty can be useful in optimization without satisfying every norm axiom.

Geometry of the unit balls

In two dimensions, $|x_1| + |x_2| = 1$ is a diamond, $x_1^2 + x_2^2 = 1$ is a circle, and $\max(|x_1|, |x_2|) = 1$ is the boundary of a square. A unit ball includes the interior, so it is defined using ≤ 1 rather than equality.

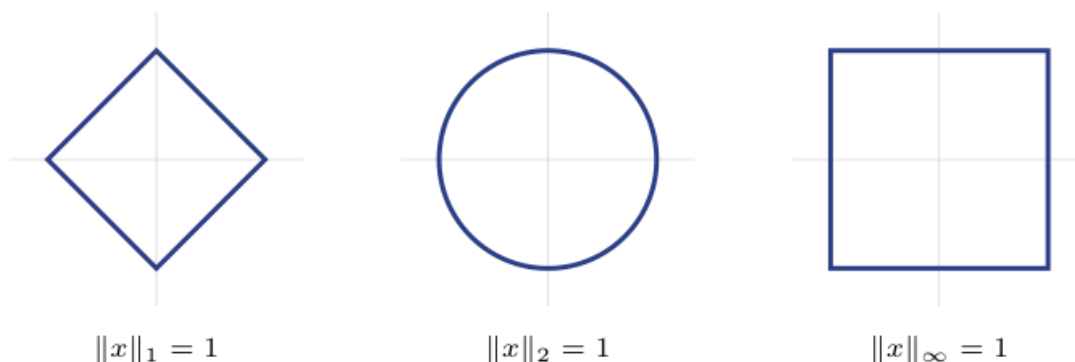


Figure 2: Boundaries of the 1-, 2-, and infinity-norm unit balls, drawn at the same scale.

The inequality $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ gives the inclusions $B_1 \subseteq B_2 \subseteq B_\infty$. A larger numerical norm gives a smaller unit ball. Ask students to test the point $(1, 1)^T$: it belongs to the infinity-norm unit ball, but not to the other two unit balls.

Bounding inner products

Part 2 slides 8–10.

Hölder's inequality states

$$|x^T y| \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad 1 \leq p, q \leq \infty.$$

The endpoint convention is $1/\infty = 0$. The case $p = q = 2$ is Cauchy–Schwarz. These bounds are useful when the exact inner product contains signs and cancellations that are inconvenient to track.

Example 12: an exact value and two bounds

For $x = (1, -2, 2)^T$ and $y = (2, 1, 2)^T$,

$$x^T y = 1 \cdot 2 + (-2) \cdot 1 + 2 \cdot 2 = 4.$$

Both Euclidean norms equal $\sqrt{9} = 3$, so Cauchy–Schwarz gives $4 \leq 9$. Also $\|x\|_1 = 5$ and $\|y\|_\infty = 2$, so the $1, \infty$ version gives $4 \leq 10$.

Neither bound is exact for this pair. To attain the $1, \infty$ bound with the same x , choose $z = (1, -1, 1)^T$. Its signs match the signs of x , and

$$x^T z = 1 + 2 + 2 = 5 = \|x\|_1 \|z\|_\infty.$$

Explain that an upper bound must be valid for every allowed vector. Equality at some vector establishes that a constant cannot be reduced uniformly; it does not require equality for the vector we happen to measure.

Cauchy–Schwarz from projection

If $y = 0$, the inequality is immediate. For $y \neq 0$ and a real scalar t ,

$$0 \leq \|x - ty\|_2^2 = \|x\|_2^2 - 2t(x^T y) + t^2 \|y\|_2^2.$$

The minimizing scalar is $t_* = (x^T y)/\|y\|_2^2$, the same scalar used in the projection example. Substitution gives

$$0 \leq \|x\|_2^2 - \frac{(x^T y)^2}{\|y\|_2^2}.$$

Multiplying by the positive denominator and taking square roots yields

$$|x^T y| \leq \|x\|_2 \|y\|_2.$$

For $y \neq 0$, equality occurs exactly when the projection residual $x - t_* y$ is zero. Equivalently, the two vectors are linearly dependent. The zero-vector cases also give equality and should be treated without dividing by zero.

Board sequence. Write the quadratic, ask students for its minimizing t , substitute it, and then identify the residual geometrically. This proves the inequality rather than using an angle formula that may itself depend on the inequality.

Norm equivalence and dimension factors

Part 2 slides 11–15.

In a fixed finite-dimensional vector space, any two norms satisfy inequalities with positive constants independent of the vector. Thus convergence to zero in one norm implies convergence in the other. The constants may depend on the dimension and on the norms themselves.

For the familiar norms,

$$\begin{aligned}\|x\|_2 &\leq \|x\|_1 \leq \sqrt{n} \|x\|_2, \\ \|x\|_\infty &\leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty, \\ \|x\|_\infty &\leq \|x\|_1 \leq n \|x\|_\infty.\end{aligned}$$

Derive the first pair in detail. Expanding the square gives

$$\|x\|_1^2 = \sum_i |x_i|^2 + 2 \sum_{i < j} |x_i| |x_j| \geq \|x\|_2^2.$$

For the upper bound, apply Cauchy–Schwarz to $(|x_i|)$ and the vector of ones:

$$\sum_i |x_i| \leq \sqrt{\sum_i |x_i|^2} \sqrt{\sum_i 1^2} = \sqrt{n} \|x\|_2.$$

For the infinity-norm comparison, the largest squared entry is at most the sum of squares, while each squared entry is at most $\|x\|_\infty^2$. Thus

$$\|x\|_\infty^2 \leq \sum_i |x_i|^2 \leq n \|x\|_\infty^2.$$

The remaining comparison follows by bounding each absolute entry by the maximum. A coordinate vector attains the lower equalities. A vector whose entries all have the same nonzero magnitude attains the upper dimension factors. These examples show that the constants are sharp.

Example 13: the same errors in a large data set

Suppose $n = 10,000$ and each error has magnitude 10^{-3} . Then

$$\|e\|_1 = 10,000 \times 10^{-3} = 10,$$

$$\|e\|_2 = \sqrt{10,000 \times 10^{-6}} = 0.1, \quad \|e\|_\infty = 10^{-3}.$$

These are three measurements of the same vector. The mean absolute error and root mean squared error include normalizations:

$$\text{MAE} = \frac{\|e\|_1}{n} = 10^{-3}, \quad \text{RMSE} = \frac{\|e\|_2}{\sqrt{n}} = 10^{-3}.$$

The equality of MAE and RMSE here follows from the equal magnitudes. It does not hold for arbitrary data. For errors $(0, 2)$, for example, MAE is 1 while RMSE is $\sqrt{2}$.

Period 1 check. The implication $\|e\|_\infty \leq 10^{-3} \Rightarrow \|e\|_2 \leq 10^{-3}$ is false in this dimension. The available bound is $\|e\|_2 \leq \sqrt{n} 10^{-3} = 0.1$. To guarantee a Euclidean error at most 10^{-3} using only a maximum-entry bound, one sufficient requirement is $\|e\|_\infty \leq 10^{-5}$.

Absolute, relative, and componentwise error

Part 2 slides 17–19.

For a reference x and an approximation $\hat{x}$, absolute error is $\|\hat{x} - x\|$. If $x \neq 0$, normwise relative error is $\|\hat{x} - x\|/\|x\|$. The numerator measures the discrepancy and the denominator fixes the scale. Both use the same chosen norm.

For the textbook example,

$$x = \begin{bmatrix} 1.234 \\ 0.05674 \end{bmatrix}, \quad \hat{x} = \begin{bmatrix} 1.235 \\ 0.05128 \end{bmatrix}, \quad \hat{x} - x = \begin{bmatrix} 0.001 \\ -0.00546 \end{bmatrix}.$$

The infinity-norm relative error is

$$\frac{0.00546}{1.234} \approx 0.004425 = 0.4425\%.$$

The largest error is in the second component, while the norm of the reference is determined by the first. Componentwise relative errors answer a different question:

$$\frac{0.001}{1.234} \approx 0.0810\%, \quad \frac{0.00546}{0.05674} \approx 9.62\%.$$

A single global percentage therefore does not certify the same accuracy in every component. Significant-digit language is only a rough shorthand and should not replace these calculations.

Example 15: a small component can be completely wrong

Let $x = (10^6, 1)^T$ and $\hat{x} = (10^6, 0)^T$. Then $\|\hat{x} - x\|_\infty = 1$, so the global relative error is 10^{-6} . The second component has relative error $|0 - 1|/1 = 100\%$.

Whether that component is scientifically important depends on the application. A small signal can still carry a quantity of interest. Do not infer importance solely from its contribution to a global norm.

If the reference vector is zero, relative error is undefined. Likewise, a componentwise relative error is undefined at a zero reference component. A mixed tolerance can handle small reference values:

$$\|\hat{x} - x\| \leq \text{atol} + \text{rtol} \|x\|.$$

The absolute tolerance has the units of the quantity. The relative tolerance is dimensionless. In an actual solver, the exact error is usually unknown, so one needs a computable estimator or a residual together with a stability bound. Part 3 explains that distinction.

Convergence and orthogonal transformations

Part 2 slides 20–22.

In the fixed space $\mathbb{R}^2$, take $x^{(k)} = (1/k, (-1)^k/k)^T$. Its norms are $2/k$, $\sqrt{2}/k$, and $1/k$. All approach zero. Alternating sign does not prevent convergence when the amplitude shrinks to zero.

Now consider the changing-dimensional vectors $z^{(n)} = (1, \dots, 1)^T/\sqrt{n} \in \mathbb{R}^n$. Their infinity-norms approach zero, but their Euclidean norms equal one. This does not contradict norm equivalence, because the vector space and its comparison constants change with n . In numerical PDEs, a continuous L^2 norm is usually approximated by a weighted sum such as $\sqrt{h^d \sum_i |x_i|^2}$ on a uniform d -dimensional grid. The weights account for cell volumes and should not be omitted when comparing resolutions.

For a square orthogonal matrix Q ,

$$\|Qx\|_2^2 = (Qx)^T(Qx) = x^T Q^T Q x = x^T x = \|x\|_2^2.$$

This identity follows entirely from $Q^T Q = I$.

Example 14: a rotation and three norms

Take

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad x = e_1.$$

Then $Qx = (1, 1)^T / \sqrt{2}$. The input has all three norms equal to one. The output has 1-norm $\sqrt{2}$, 2-norm 1, and infinity-norm $1/\sqrt{2}$. Euclidean length is unchanged, while the coordinate-dependent measurements change.

An exact orthogonal transformation preserves an existing error: $\|Qe\|_2 = \|e\|_2$. A computed product can be written $\widehat{Qx} = Qx + \delta$, where δ is newly introduced rounding error. Norm preservation is useful in stability analysis, but controlling δ requires an arithmetic error argument as well.

Eigenvectors as useful coordinates

Part 2 slides 23–28. This material comes from Section 2.1.

An eigenvector is a nonzero vector v satisfying $Hv = \lambda v$. Its direction is preserved, with scaling factor λ . The eigenvalues of an $n \times n$ matrix are the roots of $\det(H - \lambda I)$, counted over the complex numbers with algebraic multiplicity.

If n independent eigenvectors exist, placing them in a matrix X gives $HX = X\Lambda$ and $X^{-1}HX = \Lambda$. This changes the coordinates used to describe the same linear map. A general matrix need not possess enough independent eigenvectors. Real symmetric matrices do: there exists an orthogonal Q such that $Q^T H Q = \Lambda$, with real diagonal entries.

Example 9: diagonalization by hand

Let

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its characteristic polynomial is

$$\det(H - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

For $\lambda = 3$, the equations are $-x_1 + x_2 = 0$ and $x_1 - x_2 = 0$, so eigenvectors have the form $(t, t)^T$. For $\lambda = 1$, the equations give $x_1 + x_2 = 0$, so eigenvectors have the form $(t, -t)^T$.

Normalize to obtain

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad Q = [q_1 \ q_2].$$

The columns are orthonormal, and $Hq_1 = 3q_1$, $Hq_2 = q_2$. Hence

$$Q^T H Q = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = 3q_1 q_1^T + q_2 q_2^T.$$

Expanding the outer products gives

$$\frac{3}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

This reconstructs the original matrix and verifies the complete calculation. The first eigenvector is stretched by three and the perpendicular eigenvector by one.

The Rayleigh quotient as a weighted average

For a real symmetric H and nonzero x , define

$$R_H(x) = \frac{x^T H x}{x^T x}.$$

Replacing x by a nonzero scalar multiple leaves the quotient unchanged. Write $x = Qc$ in an orthonormal eigenbasis. Then

$$x^T H x = c^T \Lambda c = \sum_i \lambda_i c_i^2, \quad x^T x = \sum_i c_i^2.$$

Consequently,

$$R_H(x) = \sum_i w_i \lambda_i, \quad w_i = \frac{c_i^2}{\sum_j c_j^2}, \quad w_i \geq 0, \quad \sum_i w_i = 1.$$

A weighted average lies between the smallest and largest values averaged. Therefore

$$\lambda_{\min}(H) \leq R_H(x) \leq \lambda_{\max}(H).$$

Taking x to be an extreme eigenvector attains the corresponding bound. Thus the minimum and maximum are exactly the extreme eigenvalues. Multiplying by $\|x\|_2^2$ gives the quadratic-form inequality

$$\lambda_{\min}(H) \|x\|_2^2 \leq x^T H x \leq \lambda_{\max}(H) \|x\|_2^2.$$

Where symmetry enters. It supplies a real orthonormal eigenbasis. Without that basis, the denominator need not be a sum of squared eigen-coordinates with the same weights. Do not use this proof for an arbitrary nonsymmetric matrix.

Example 9 continued: the curve and a mixed direction

For $x(\theta) = \cos \theta q_1 + \sin \theta q_2$, the norm equals one and

$$R_H(x(\theta)) = 3 \cos^2 \theta + \sin^2 \theta.$$

The maximum is three when the vector lies along q_1 , and the minimum is one when it lies along q_2 . Changing the sign of a vector does not change its quotient.

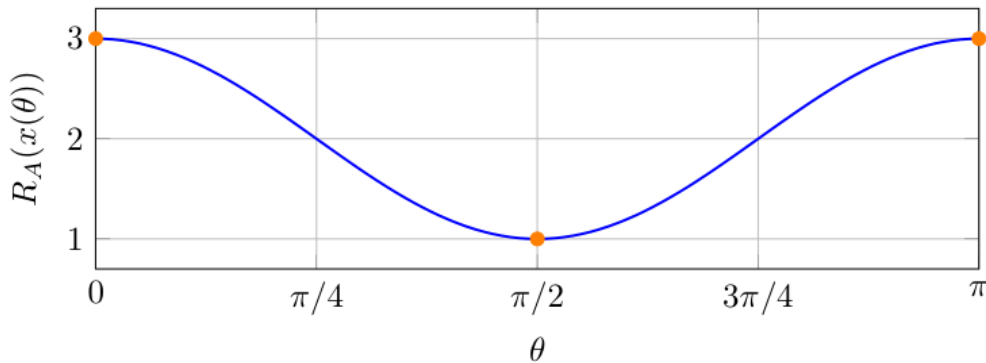


Figure 3: Rayleigh quotient as the unit input rotates in the eigenbasis of the symmetric example.

For the paired calculation on slide 30, take $x = (2, 1)^T$. Directly,

$$Hx = (5, 4)^T, \quad x^T Hx = 2 \cdot 5 + 1 \cdot 4 = 14, \quad x^T x = 5.$$

Thus $R_H(x) = 14/5 = 2.8$. The eigen-coordinates are

$$c_1 = q_1^T x = 3/\sqrt{2}, \quad c_2 = q_2^T x = 1/\sqrt{2}.$$

Their squared values are $9/2$ and $1/2$, giving weights $9/10$ and $1/10$. The weighted average is $3(9/10) + 1(1/10) = 2.8$, agreeing with direct multiplication.

Period 2 practice and answers

Part 2 slides 31–33.

For $x = (100, 1)^T$ and $\hat{x} = (100, 0)^T$, the error is $(0, -1)^T$. The global infinity-norm relative error is $1/100 = 1\%$, while the second component's relative error is $1/1 = 100\%$.

For $H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $R_H(e_1) = 2$. A unit vector attaining the maximum is $(1, 1)^T/\sqrt{2}$, with quotient three. Students should state that H is symmetric when invoking the extreme-eigenvalue characterization.

The exit questions test the failed homogeneity of squared length, the fixed-dimensional meaning of norm equivalence, and the symmetry assumption behind orthogonal diagonalization. Before Part 3, ask students to form $M^T M$ for $M = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix}$, keeping the off-diagonal sign.

Optional study for Part 2

Part 2 slides 34–39. This material is outside the scheduled lesson.

Example 8: repeated eigenvalues and complex eigenvalues

For $J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the characteristic polynomial is $(1 - \lambda)^2$. The eigenvalue one has algebraic multiplicity two. However,

$$(J - I)x = \begin{bmatrix} x_2 \\ 0 \end{bmatrix} = 0$$

requires $x_2 = 0$, so the eigenspace is only one-dimensional. No basis of two independent eigenvectors exists. Repetition alone does not force this failure: the identity also has a repeated eigenvalue but every nonzero vector is an eigenvector.

For $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, the characteristic polynomial is $\lambda^2 + 1$, so the eigenvalues are i and $-i$. Over the real plane this matrix rotates every vector through 90 degrees and preserves no nonzero real line. Real matrix entries do not require real eigenvalues.

Weighted norms and invisible directions

Define $\|x\|_C = \|Cx\|_2$ for $C \in \mathbb{R}^{m \times n}$. Homogeneity and the triangle inequality follow from the Euclidean norm. Positivity holds exactly when $Cx = 0$ implies $x = 0$, which means that C has full column rank.

$$\text{For } C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix},$$

$$\|x\|_C = \sqrt{x_1^2 + 4x_2^2}.$$

The matrix is rectangular, but the expression is a norm. For $D = \text{diag}(1, 0)$, $\|De_2\|_2 = 0$ even though $e_2 \neq 0$. The resulting expression is a seminorm. This connects directly to the null-space interpretation from Part 1.

A proof of general finite-dimensional norm equivalence

For any norm f and the coordinate basis, the triangle inequality gives

$$f(x) = f\left(\sum_i x_i e_i\right) \leq \sum_i |x_i| f(e_i) \leq C \|x\|_2, \quad C = \sqrt{\sum_i f(e_i)^2}.$$

The reverse triangle inequality gives $|f(x) - f(y)| \leq f(x - y) \leq C \|x - y\|_2$, so f is continuous in Euclidean coordinates. On the Euclidean unit sphere, compactness gives a minimum $c > 0$. Positivity ensures it cannot be zero. By homogeneity, $c \|x\|_2 \leq f(x) \leq C \|x\|_2$ for every x . Comparing two norms through the Euclidean norm proves their equivalence.

This proof explains why the constants exist. The explicit 1-, 2-, and infinity-norm inequalities are more useful when an actual numerical tolerance must be converted.

Part 3. Matrix norms, iterative solves, and perturbations

Learning outcomes and teaching plan

Students should be able to calculate the 1-, infinity-, Frobenius, and spectral norms of a small matrix; derive the induced-norm product bound; compute several steps of power and fixed-point iterations; and use a Neumann bound to quantify inverse perturbation. They should distinguish a residual from the solution error and explain the role of the inverse in connecting them.

Period 1: measuring a linear map.

Minutes	Slides	Teaching and student work
0–4	1–2	Recall the Rayleigh quotient and state the amplification question.
4–12	3–6	Entrywise and induced norms, action bound, product proof.
12–23	7–10	Row/column formulas, attaining vectors, and proofs.
23–28	11–12	Frobenius norm and inexpensive spectral bounds.
28–40	13–16	Spectral norm, exact example, attaining direction, ellipse.
40–45	17	Independent four-norm calculation.

Slide 18 marks the break. **Period 2: iteration and sensitivity.**

Minutes	Slides	Teaching and student work
0–5	19–20	Discuss the calculation and the eigenvalue/norm counterexample.
5–13	21–22	Power iteration and four hand-checkable iterates.
13–25	23–26	Neumann proof, truncation error, and fixed-point iterates.
25–35	27–28	Inverse perturbation: factor order, bound, exact comparison.
35–42	29–30	Small residual with large error; paired interpretation.
42–45	31	Exit questions and the link to conditioning.

A matrix norm as a worst-case amplification

Part 3 slides 3–6.

Suggested opening. “A vector norm measures the size of an input or an error. A matrix norm can measure how much a linear map amplifies that input. We will use this interpretation to decide whether repeated corrections shrink an error and whether a small residual implies a small solution error.”

Treating an $m \times n$ matrix as a vector of mn entries gives the Frobenius norm:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}.$$

It is the total Euclidean size of the entries. An induced p -norm instead measures the largest ratio between output and input sizes:

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_p.$$

The two expressions agree because a nonzero input can be divided by its norm. In finite dimension the unit sphere is compact and the output norm is continuous, so the maximum is attained.

For every $x \neq 0$, the ratio in the definition is at most the supremum. Multiplying by $\|x\|_p$ gives

$$\|Ax\|_p \leq \|A\|_p \|x\|_p.$$

For $x = 0$, the same inequality is immediate. Thus it holds for all inputs, including an error vector.

Submultiplicativity, with the intermediate space visible

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times r}$. For an input $x \in \mathbb{R}^r$,

$$\|ABx\|_p \leq \|A\|_p \|Bx\|_p \leq \|A\|_p \|B\|_p \|x\|_p.$$

Divide by $\|x\|_p$ for nonzero x , then take the maximum to obtain

$$\|AB\|_p \leq \|A\|_p \|B\|_p.$$

The middle vector Bx belongs to $\mathbb{R}^n$. Matching the norm used for that intermediate space is essential. Equality is not required: B may send its most amplified input into a direction that A strongly contracts.

For example, $A = \text{diag}(1, 0)$ and $B = \text{diag}(0, 1)$ each have 2-norm one, while $AB = 0$. This gives a strict product inequality with an easily checked reason.

Column and row formulas, including attaining inputs

Part 3 slides 7–10.

The common induced norms have formulas

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \quad \|A\|_\infty = \max_i \sum_j |a_{ij}|.$$

The 1-norm is the maximum absolute **column** sum; the infinity-norm is the maximum absolute **row** sum. A matrix 1-norm is not the sum of the magnitudes of all its entries.

Proof of the column-sum formula

Write $Ax = \sum_j x_j a_j$. The triangle inequality gives

$$\|Ax\|_1 \leq \sum_j |x_j| \|a_j\|_1 \leq \left(\max_j \|a_j\|_1 \right) \|x\|_1.$$

This proves that the proposed column sum is an upper bound. To show it is exact, choose j_* with the largest column sum and take $x = e_{j_*}$. Then $\|x\|_1 = 1$ and $Ax = a_{j_*}$, attaining the bound.

Proof of the row-sum formula

For every row i ,

$$|(Ax)_i| \leq \sum_j |a_{ij}| |x_j| \leq \left(\sum_j |a_{ij}| \right) \|x\|_\infty.$$

Taking the maximum over rows proves the upper bound. For a row i_* with largest sum, choose $x_j = 1$ when $a_{i_*j} \geq 0$ and $x_j = -1$ when $a_{i_*j} < 0$. Then $\|x\|_\infty = 1$ and

$$(Ax)_{i_*} = \sum_j |a_{i_*j}|.$$

Assigning a sign to zero entries in this way avoids an unnecessarily zero input. The zero matrix itself can also be treated separately.

The proof pattern is worth emphasizing: first bound the output for **every** unit input, then construct **one** input attaining that bound.

Example 16: signed entries and four different norms

Use

$$M = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix}.$$

The absolute column sums are $1 + 2 = 3$ and $7 + 3 = 10$, so $\|M\|_1 = 10$. The absolute row sums are $1 + 7 = 8$ and $2 + 3 = 5$, so $\|M\|_\infty = 8$.

For the 1-norm, $x = e_2$ attains the maximum because $Me_2 = (-7, -3)^T$ has 1-norm ten. For the infinity-norm, choose $x = (1, -1)^T$, matching the signs of the first row. Then

$$Mx = \begin{bmatrix} 8 \\ 1 \end{bmatrix}, \quad \|x\|_\infty = 1, \quad \|Mx\|_\infty = 8.$$

Adding the signed row entries instead would give $1 - 7 = -6$. Its magnitude six is not the absolute row sum eight. Take absolute values before adding.

Frobenius norm and useful bounds

Part 3 slides 11–12.

For the same matrix,

$$\|M\|_F = \sqrt{1^2 + (-7)^2 + (-2)^2 + (-3)^2} = \sqrt{63} \approx 7.9373.$$

Any unit vector gives a lower bound on the spectral norm. Taking e_2 yields

$$\|M\|_2 \geq \|Me_2\|_2 = \sqrt{49 + 9} = \sqrt{58} \approx 7.6158.$$

For an upper bound, apply Cauchy–Schwarz to each row:

$$\|Ax\|_2^2 = \sum_i \left(\sum_j a_{ij}x_j \right)^2 \leq \sum_i \left(\sum_j a_{ij}^2 \right) \|x\|_2^2 = \|A\|_F^2 \|x\|_2^2.$$

Hence $\|A\|_2 \leq \|A\|_F$. For our example,

$$7.6158 \lesssim \|M\|_2 \lesssim 7.9373.$$

The bound $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty}$ gives $\sqrt{80} \approx 8.9443$ here, which is valid but less sharp. Bounds can differ in quality, and an inexpensive bound may still be useful when the exact norm is costly.

For an $m \times n$ matrix, the broader comparisons include

$$\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{m} \|A\|_\infty,$$

$$\frac{1}{\sqrt{m}} \|A\|_1 \leq \|A\|_2 \leq \sqrt{n} \|A\|_1.$$

Check which dimension belongs to the input and output. These inequalities follow by applying the vector norm comparisons before and after the matrix action.

The matrix 2-norm and the Rayleigh quotient

Part 3 slides 13–16.

For every real matrix A , the matrix $A^T A$ is symmetric and positive semidefinite because

$$x^T A^T A x = \|Ax\|_2^2 \geq 0.$$

Applying the Rayleigh characterization from Part 2 gives

$$\|A\|_2^2 = \max_{\|x\|_2=1} \|Ax\|_2^2 = \max_{\|x\|_2=1} x^T A^T A x = \lambda_{\max}(A^T A).$$

Therefore

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}.$$

An eigenvector of $A^T A$ for its largest eigenvalue, normalized to length one, attains this norm. Its direction is a right singular-vector direction of A .

Example 16 continued: the exact spectral norm

Multiply the Gram matrix entry by entry:

$$M^T M = \begin{bmatrix} 1 & -2 \\ -7 & -3 \end{bmatrix} \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ -1 & 58 \end{bmatrix}.$$

The off-diagonal entry is $1(-7) + (-2)(-3) = -7 + 6 = -1$. This sign is easy to lose.

The characteristic equation is

$$(5 - \lambda)(58 - \lambda) - 1 = \lambda^2 - 63\lambda + 289 = 0.$$

Its discriminant is $63^2 - 4 \cdot 289 = 3969 - 1156 = 2813$, giving

$$\lambda_{\pm} = \frac{63 \pm \sqrt{2813}}{2}.$$

The larger eigenvalue is approximately 58.0188612. Thus

$$\|M\|_2 = \sqrt{\frac{63 + \sqrt{2813}}{2}} \approx 7.6170113.$$

It lies between the previously computed lower and upper bounds. The close lower bound from e_2 suggests that the maximizing direction is near e_2 .

The attaining direction

Let $\lambda_+ = (63 + \sqrt{2813})/2$. The first row of $(M^T M - \lambda_+ I)w = 0$ gives

$$(5 - \lambda_+)w_1 - w_2 = 0.$$

Choose $w_1 = -1$, so $w_2 = \lambda_+ - 5$. Therefore

$$z = \frac{1}{\sqrt{1 + (\lambda_+ - 5)^2}} \begin{bmatrix} -1 \\ \lambda_+ - 5 \end{bmatrix} \approx \begin{bmatrix} -0.01886 \\ 0.99982 \end{bmatrix}.$$

The second equation also holds because λ_+ satisfies the characteristic equation. The vector is unit length and

$$\|Mz\|_2^2 = z^T M^T M z = \lambda_+ z^T z = \lambda_+.$$

Hence z attains the norm. Decimal rounding is for interpretation; use the exact eigenvalue when checking the identity.

Example 17: a circle becomes an ellipse

For $D = \text{diag}(3, 1)$ and a unit vector $x = (\cos \theta, \sin \theta)^T$,

$$Dx = (3 \cos \theta, \sin \theta)^T, \quad \|Dx\|_2^2 = 9 \cos^2 \theta + \sin^2 \theta.$$

The output curve is an ellipse with semiaxes three and one. The largest output radius is three, which is $\|D\|_2$. The Frobenius norm is $\sqrt{3^2 + 1^2} = \sqrt{10}$, which combines the squared semiaxes rather than selecting the largest.

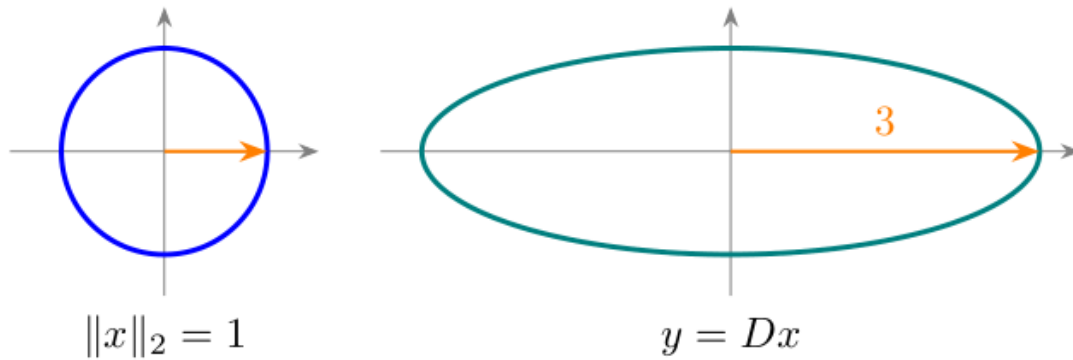


Figure 4: A unit circle mapped to an ellipse by a diagonal matrix. The same length scale is used in both panels.

For numerical work, the identity involving $A^T A$ is valuable for analysis, but explicitly forming a Gram matrix can square the 2-norm condition number when A has full column rank. A general spectral norm is normally calculated with a singular-value method or estimated by matrix-vector products. For a diagonal matrix, it is immediate from the diagonal entries.

Period 1 exercise and the restart

Part 3 slides 17 and 19.

For $B = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$, the absolute column sums are one and four, and the row sums are three and two.

Therefore

$$\|B\|_1 = 4, \quad \|B\|_\infty = 3, \quad \|B\|_F = \sqrt{1 + 4 + 4} = 3.$$

The Gram matrix is

$$B^T B = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}.$$

Its characteristic polynomial is $(1 - \lambda)(8 - \lambda) - 4 = \lambda^2 - 9\lambda + 4$. Hence

$$\|B\|_2 = \sqrt{\frac{9 + \sqrt{65}}{2}} \approx 2.9208.$$

Taking the unit vector e_2 gives the lower bound $\sqrt{8} \approx 2.8284$, and the Frobenius norm gives the upper bound three. A complete answer should identify the larger eigenvalue and explain why its square root is required.

Eigenvalues of a general matrix do not measure its largest stretch

Part 3 slide 20.

Example 18: zero eigenvalues with nonzero amplification

Let

$$N = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix}.$$

Its characteristic polynomial is λ^2 , so both eigenvalues are zero. Nevertheless, $Ne_2 = 5e_1$, and

$$N^T N = \begin{bmatrix} 0 & 0 \\ 0 & 25 \end{bmatrix}, \quad \|N\|_2 = 5.$$

The matrix is nonnormal: $N^T N \neq N N^T$. Its eigenvalues describe its invariant directions and its powers, while the spectral norm describes the largest one-step amplification of any input. Here $N^2 = 0$, so a vector can grow on the first application and vanish on the second.

For a real symmetric matrix, orthogonal diagonalization does imply $\|A\|_2 = \max_i |\lambda_i(A)|$. The symmetry assumption is what makes this special shortcut valid in the present course.

Power iteration without forming a Gram matrix

Part 3 slides 21–22.

Example 19: the algorithm

Choose a unit starting vector x_0 . At each step compute

$$y_k = Ax_k, \quad z_k = A^T y_k, \quad x_{k+1} = \frac{z_k}{\|z_k\|_2}, \quad \mu_k = \|Ax_k\|_2.$$

The update is power iteration on $A^T A$, implemented through two matrix-vector products. If $z_k = 0$, the normalization is undefined and the implementation must stop or choose another starting vector.

When the largest eigenvalue of $A^T A$ is simple and the initial vector has a nonzero component in its eigendirection, normalized powers approach that direction up to sign. Expand $x_0 = \sum_i c_i q_i$. After k unnormalized steps,

$$(A^T A)^k x_0 = \sum_i c_i \lambda_i^k q_i.$$

Dividing by the dominant factor shows that each other component is scaled relative to the first by $(\lambda_i/\lambda_1)^k$. This explains the convergence condition rather than just stating it.

Four hand-checkable iterates

For $A = \text{diag}(3, 1)$ and $x_0 = (1, 1)^T/\sqrt{2}$, the first products are

$$y_0 = \frac{1}{\sqrt{2}}(3, 1)^T, \quad z_0 = \frac{1}{\sqrt{2}}(9, 1)^T.$$

Since $\|z_0\|_2 = \sqrt{82/2}$,

$$x_1 = \frac{(9, 1)^T}{\sqrt{82}}.$$

At every step the ratio of the second coordinate to the first shrinks by $1/9$. Inductively,

$$x_k = \frac{(9^k, 1)^T}{\sqrt{9^{2k} + 1}}, \quad \mu_k = \sqrt{\frac{9 \cdot 9^{2k} + 1}{9^{2k} + 1}}.$$

k	$(x_k)_1$	$(x_k)_2$	μ_k
0	0.707107	0.707107	2.236068
1	0.993884	0.110432	2.983696
2	0.999924	0.012345	2.999797
3	0.999999	0.001372	2.999997

Every μ_k is a lower bound on the true spectral norm because x_k has unit length. A nearly unchanged estimate is not by itself a universal error certificate. If we start exactly at e_2 in this example, the iteration remains there and reports one forever, missing the dominant direction.

The Neumann series and an error bound

Part 3 slides 23–26.

For a scalar $|f| < 1$, the geometric series sums to $(1 - f)^{-1}$. For matrices, an induced norm gives a sufficient condition that controls every vector direction simultaneously. Assume

$$q = \|F\|_p < 1.$$

First prove $I - F$ is nonsingular. If $(I - F)x = 0$ for a nonzero x , then $x = Fx$, so

$$\|x\|_p = \|Fx\|_p \leq q\|x\|_p < \|x\|_p,$$

a contradiction.

Define the finite sum $S_N = I + F + \cdots + F^N$. Multiplication telescopes:

$$S_N(I - F) = I - F^{N+1}.$$

By submultiplicativity, $\|F^{N+1}\|_p \leq q^{N+1} \rightarrow 0$. Thus S_N converges to $(I - F)^{-1}$, and

$$\|(I - F)^{-1}\|_p \leq \sum_{k=0}^{\infty} q^k = \frac{1}{1 - q}.$$

The remainder is bounded by the tail of the scalar series:

$$\|(I - F)^{-1} - S_N\|_p \leq \sum_{k=N+1}^{\infty} q^k = \frac{q^{N+1}}{1 - q}.$$

The bound converts convergence into a quantitative stopping rule. For $0 < q < 1$ and a desired error at most ε , choose a nonnegative integer N with $q^{N+1} \leq \varepsilon(1 - q)$. Because $\log q < 0$, dividing a logarithmic inequality reverses its direction. If $q = 0$, then $F = 0$ and the inverse is already I .

Example 20: four terms and the exact error

Let $F = \text{diag}(1/2, 1/4)$. Its infinity-norm is $q = 1/2$, and

$$(I - F)^{-1} = \text{diag}(2, 4/3).$$

Using four terms means S_3 , whose highest power is three:

$$\begin{aligned} S_3 &= I + F + F^2 + F^3 \\ &= \text{diag}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}, 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}\right) \\ &= \text{diag}(15/8, 85/64). \end{aligned}$$

The first diagonal error is $2 - 15/8 = 1/8$. The second is

$$\frac{4}{3} - \frac{85}{64} = \frac{256 - 255}{192} = \frac{1}{192}.$$

Therefore the infinity-norm error is $1/8$. The bound predicts

$$\frac{(1/2)^4}{1 - 1/2} = \frac{1}{8},$$

which is exactly attained. To make this operator error at most 10^{-3} , the bound becomes $2^{-N} \leq 10^{-3}$. Taking $N = 10$ works, meaning **11 terms**, from I through F^{10} .

The same series as a fixed-point solver

To solve $(I - F)x = b$, write $x = b + Fx$ and use

$$x_{k+1} = b + Fx_k, \quad x_0 = 0.$$

The first iterates are $x_1 = b$, $x_2 = b + Fb$, and $x_3 = b + Fb + F^2b$. Thus $x_k = S_{k-1}b$ for $k \geq 1$.

For $b = (1, 1)^T$ and the diagonal F above:

k	x_k	$\ x_* - x_k\ _\infty$
0	$(0, 0)^T$	2
1	$(1, 1)^T$	1
2	$(3/2, 5/4)^T$	$1/2$
3	$(7/4, 21/16)^T$	$1/4$
4	$(15/8, 85/64)^T$	$1/8$

The exact solution is $x_* = (2, 4/3)^T$. Subtracting the fixed-point equation from the update gives $e_{k+1} = Fe_k$, whether we consistently define the error as $x_k - x_*$ or its negative. Hence

$$\|e_{k+1}\|_p \leq q\|e_k\|_p.$$

For this example the slower first coordinate determines the maximum error. This is the simplest demonstration of an iteration whose different directions converge at different speeds.

Perturbing a nonsingular matrix

Part 3 slides 27–28.

Suppose A is nonsingular and E is a perturbation. The relevant small quantity is the perturbation measured relative to the action of the inverse:

$$G = A^{-1}E, \quad r = \|G\|_p < 1.$$

Then

$$A + E = A(I + G), \quad (A + E)^{-1} = (I + G)^{-1}A^{-1}.$$

Multiplication order matters. The first identity is checked by $A(I + A^{-1}E) = A + E$. In general, replacing $A^{-1}E$ by EA^{-1} changes the quantity appearing in the hypothesis.

Subtract the original inverse:

$$\begin{aligned} (A + E)^{-1} - A^{-1} &= [(I + G)^{-1} - I]A^{-1} \\ &= -(I + G)^{-1}GA^{-1}. \end{aligned}$$

Apply the Neumann result with $F = -G$. Because $\| -G \|_p = r < 1$,

$$\|(I + G)^{-1}\|_p \leq \frac{1}{1 - r}.$$

Submultiplicativity then gives

$$\|(A + E)^{-1} - A^{-1}\|_p \leq \frac{\|G\|_p \|A^{-1}\|_p}{1 - r} \leq \frac{\|E\|_p \|A^{-1}\|_p^2}{1 - r}.$$

This both proves invertibility of the perturbed matrix and bounds the change in its inverse. If $r \geq 1$, this theorem does not apply; it does not prove that the perturbed matrix is singular.

Example 21: evaluating every term

Take

$$A = \text{diag}(2, 1), \quad E = \text{diag}(0, -1/10).$$

Then

$$A^{-1} = \text{diag}(1/2, 1), \quad G = A^{-1}E = \text{diag}(0, -1/10).$$

Thus $r = 1/10 < 1$, $\|E\|_\infty = 1/10$, and $\|A^{-1}\|_\infty = 1$. The perturbed matrix and inverse are

$$A + E = \text{diag}(2, 9/10), \quad (A + E)^{-1} = \text{diag}(1/2, 10/9).$$

The exact difference is $\text{diag}(0, 1/9)$, with infinity-norm $1/9$. The bound gives

$$\frac{(1/10) \cdot 1^2}{1 - 1/10} = \frac{1}{9}.$$

The bound is sharp for this example. The change decreases the smaller diagonal entry and therefore increases its inverse response.

Why a residual needs a conditioning interpretation

Part 3 slides 29–31.

For a computed vector $\hat{x}$, the residual is $r = b - A\hat{x}$. It measures how well the computed vector satisfies the equation. If $Ax_* = b$, then

$$A(x_* - \hat{x}) = r, \quad x_* - \hat{x} = A^{-1}r.$$

This is the exact connection to the forward solution error. Taking an induced norm gives

$$\|x_* - \hat{x}\| \leq \|A^{-1}\| \|r\|.$$

If $b \neq 0$, then $x_* \neq 0$ and $\|b\| \leq \|A\| \|x_*\|$. Combining these inequalities yields

$$\frac{\|x_* - \hat{x}\|}{\|x_*\|} \leq \underbrace{\|A\| \|A^{-1}\|}_{\kappa(A)} \frac{\|r\|}{\|b\|}.$$

This is a useful preview of condition numbers. It is a worst-case upper bound, so it can be conservative for a particular residual direction.

A small residual with a large forward error

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 10^{-4} \end{bmatrix}, \quad x_* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 10^{-4} \end{bmatrix}, \quad \hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The residual is $(0, 10^{-4})^T$, giving

$$\frac{\|r\|_2}{\|b\|_2} = \frac{10^{-4}}{\sqrt{1 + 10^{-8}}} \approx 10^{-4}.$$

The solution error is $(0, 1)^T$, so

$$\frac{\|x_* - \hat{x}\|_2}{\|x_*\|_2} = \frac{1}{\sqrt{2}} \approx 70.7\%.$$

The second coordinate is weakly visible in the residual because A multiplies it by 10^{-4} . The inverse multiplies that residual coordinate by 10^4 , restoring the order-one error. Here $\kappa_2(A) = 10^4$, consistent with a relative-error upper bound near one.

Paired discussion. Ask which component is poorly observed, which inverse entry amplifies it, and what additional estimate is required to certify solution accuracy. The answer is not to discard residuals. Residuals are computable and useful, but they should be interpreted using the stability or conditioning of the underlying equation. The same principle applies when a learned model is assessed using a discretized PDE residual.

Optional study for Part 3

Part 3 slides 32–49. This material is outside the scheduled lesson.

Entrywise maximum and mixed induced norms

Let $\|A\|_{\max} = \max_{i,j} |a_{ij}|$. This satisfies the vector-space norm axioms, but not the same-norm submultiplicative inequality in general. For the 2×2 all-ones matrix J , $\|J\|_{\max} = 1$, while $J^2 = 2J$ and $\|J^2\|_{\max} = 2$.

The entrywise maximum nevertheless has a precise operator interpretation when the input uses the 1-norm and the output uses the infinity-norm:

$$\|A\|_{1 \rightarrow \infty} = \max_{\|x\|_1=1} \|Ax\|_{\infty} = \max_{i,j} |a_{ij}|.$$

If $M = \max_{i,j} |a_{ij}|$, then $|(Ax)_i| \leq M \sum_j |x_j| = M\|x\|_1$. Taking a coordinate vector at a column containing a largest entry attains M . Composition inequalities work when the output norm of the first map matches the input norm of the next. They need not use the same mixed norm for both factors.

Frobenius products and submatrices

For compatible matrices, columnwise application of the spectral action bound gives

$$\|AB\|_F^2 = \sum_j \|AB(:, j)\|_2^2 \leq \|A\|_2^2 \|B\|_F^2 \leq \|A\|_F^2 \|B\|_F^2.$$

So the Frobenius norm is submultiplicative. It is not an induced norm from the same vector norm on a square space when $n > 1$, because any such induced norm has $\|I_n\| = 1$, whereas $\|I_n\|_F = \sqrt{n}$.

If a submatrix is written $B = RAC$ with coordinate selection and insertion matrices, then $\|R\|_p = \|C\|_p = 1$ for nonempty selections in the usual coordinate p -norms. Hence $\|B\|_p \leq \|A\|_p$. The argument does not automatically extend to arbitrary weighted norms, where coordinate selection can have a different norm.

For dense matrices, the 1-, infinity-, and Frobenius norms require one pass through the entries, costing $O(mn)$. Sparse versions depend on the stored nonzero entries. A general spectral norm requires additional work, although special matrices can be handled directly.

Norm conditions, invertibility, and series convergence

For $F = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$, the 2-norm is two, but $F^2 = 0$. The Neumann series terminates:

$$(I - F)^{-1} = I + F = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Thus $\|F\|_2 < 1$ is not necessary for convergence. Conversely, $F = 2I$ gives an invertible $I - F = -I$, while its Neumann series diverges. Invertibility alone is not sufficient for this particular series representation. Later eigenvalue theory gives the exact convergence criterion $\rho(F) < 1$ for a fixed square matrix, where ρ is its spectral radius.

Example 22: orthogonal invariance for matrices

If Q and Z are square orthogonal matrices of compatible sizes, then

$$\|QAZ\|_2 = \|A\|_2, \quad \|QAZ\|_F = \|A\|_F.$$

For the spectral norm, left multiplication preserves output Euclidean length, and right multiplication carries the unit input sphere onto itself. For the Frobenius norm, sum the squared Euclidean lengths of the columns to prove left invariance. Right invariance follows by transposition and the same argument.

Take $A = \text{diag}(3, 1)$ and $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$. Then

$$QA = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}, \quad (QA)^T(QA) = \text{diag}(9, 1).$$

Thus $\|QA\|_2 = 3$ and $\|QA\|_F = \sqrt{(9 + 1 + 9 + 1)/2} = \sqrt{10}$. Its maximum column sum is $3\sqrt{2}$, and its maximum row sum is $4/\sqrt{2} = 2\sqrt{2}$. The original matrix has both of those norms equal to three. Orthogonal invariance is specific to the norms in the theorem.

The norm of a rank-one matrix

For $E = uv^T$ and a unit Euclidean vector x ,

$$\|Ex\|_2 = \|u\|_2 |v^T x| \leq \|u\|_2 \|v\|_2.$$

If $v \neq 0$, choose $x = v/\|v\|_2$ to attain the bound. Also,

$$\|uv^T\|_F^2 = \sum_{i,j} u_i^2 v_j^2 = \left(\sum_i u_i^2 \right) \left(\sum_j v_j^2 \right).$$

Therefore $\|uv^T\|_2 = \|uv^T\|_F = \|u\|_2 \|v\|_2$. The identity also holds when either vector is zero, although the normalized attaining-vector expression then needs separate handling.

Example 23: a minimum-norm correction satisfying one observation

Suppose we want $(A + E)s = y$ for a given nonzero s . Put $r = y - As$, so the constraint is $Es = r$. Every feasible correction obeys

$$\|E\|_2 \geq \frac{\|Es\|_2}{\|s\|_2} = \frac{\|r\|_2}{\|s\|_2}.$$

Choose

$$E_* = \frac{rs^T}{s^T s}.$$

Then $E_*s = r$ and the rank-one norm formula gives $\|E_*\|_2 = \|r\|_2/\|s\|_2$. A lower bound valid for every feasible correction, together with a correction attaining it, proves optimality.

For $A = I_2$, $s = (1, 2)^T$, and $y = (3, 1)^T$, we have $r = (2, -1)^T$ and $s^T s = 5$. Hence

$$E_* = \frac{1}{5} \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}, \quad E_*s = \frac{1}{5} \begin{bmatrix} 10 \\ -5 \end{bmatrix} = r.$$

Therefore $(I + E_*)s = y$. Since $\|r\|_2 = \|s\|_2 = \sqrt{5}$, the spectral and Frobenius norms of the correction both equal one.

The spectral-norm minimizer need not be unique. For the Frobenius norm, this correction is the unique minimizer: if $E = E_* + D$ is another feasible correction, then $Ds = 0$ and

$$\langle E_*, D \rangle_F = \frac{r^T Ds}{s^T s} = 0.$$

Consequently $\|E\|_F^2 = \|E_*\|_F^2 + \|D\|_F^2$, with equality only when $D = 0$. The constraint specifies the action on one input vector; it does not determine the desired action on every input.

Optional computational workshop

This workshop is homework or tutorial material, additional to the three scheduled sessions. It reinforces the hand calculations without requiring students to implement a factorization algorithm before that topic has been taught.

Reusing a solve in a Woodbury update

The following function accepts a callable `solve_A`. It should apply a previously constructed solver or factorization to one or several right-hand sides. Passing `numpy.linalg.solve(A, rhs)` through a wrapper is sufficient for checking the tiny example, but it does not itself demonstrate reuse of an existing factorization.

```

import numpy as np

def woodbury_solve(solve_A, U, V, b):
    y = solve_A(b)
    W = solve_A(U)
    S = np.eye(U.shape[1]) + V.T @ W
    c = np.linalg.solve(S, V.T @ y)
    return y - W @ c

U = np.array([[1., 0.], [0., 1.], [1., 1.]])
b = np.array([1., 1., 0.])
x = woodbury_solve(lambda rhs: rhs / 2, U, U, b)
residual = b - (2 * np.eye(3) + U @ U.T) @ x

```

For this example, the result is $(0.4, 0.4, -0.2)^T$, with a residual at floating-point rounding level. Ask students to identify the sizes of W and S , and to explain why the returned residual uses the updated matrix. The function assumes nonsingular systems. A practical implementation should treat small or ill-conditioned update systems carefully; detecting a singular system exception alone does not estimate conditioning.

Spectral norm estimation by products

```

def spectral_norm_iterations(A, x, steps=4):
    x = np.asarray(x, dtype=float)
    length = np.linalg.norm(x)
    if length == 0:
        raise ValueError("The starting vector must be nonzero.")
    x = x / length
    estimates = []
    for _ in range(steps):
        y = A @ x
        estimates.append(np.linalg.norm(y))
        z = A.T @ y
        length = np.linalg.norm(z)
        if length == 0:
            break
        x = z / length
    return np.array(estimates)

```

Use $A = \text{diag}(3, 1)$ and starting vector $(1, 1)^T$ to recover the table in Example 19. Then start from $(0, 1)^T$ and explain why all estimates equal one. The purpose is to test the hypothesis about the initial direction, not merely to obtain a small stopping difference.

Questions that connect the three sessions

1. In the running rank-two example, why does adding a null vector change the parameters without changing the output? Which object in a prediction problem would play the role of this matrix?
2. Compare the errors $(1, 0, \dots, 0)^T$ and $(1/\sqrt{n}, \dots, 1/\sqrt{n})^T$. Both have Euclidean norm one. How do their infinity-norms differ, and why might a maximum-entry error matter?

3. Replace $\text{diag}(1, 10^{-4})$ in the residual example by $\text{diag}(1, \delta)$, with $0 < \delta \leq 1$. Derive the residual and relative solution error for the same x_* and $\hat{x}$. Explain what changes as δ decreases.

Answers. First, the added vector is annihilated by the map, so the output is unchanged. A design matrix in linear regression is one example, with its null space describing parameter directions invisible to those observations. Second, the maximum-entry errors are one and $1/\sqrt{n}$; a global Euclidean metric does not identify whether error is concentrated or spread out. Third, the residual is $(0, \delta)^T$ and its relative norm is $\delta/\sqrt{1 + \delta^2}$, while relative solution error remains $1/\sqrt{2}$. The condition number is $1/\delta$, so a smaller residual reflects weaker sensitivity of the equation to the second component rather than improved recovery of that component.

References and reading

The main reference is G. H. Golub and C. F. Van Loan, *Matrix Computations*, fourth edition, Johns Hopkins University Press, 2013, Sections 2.1–2.3, pp. 63–75. The supplied excerpt and the expanded slides determine the scope. Original theorem and equation numbers are retained in the slides; the added example numbers refer to the expanded teaching deck.

Chinese companion textbook: Wang Lei, *Juzhen Lilun yu Jisuan* (Matrix Theory and Computation). Use the matching chapters for terminology and additional practice.

Recommended reading order is Section 2.1 for Part 1, Section 2.2 followed by the symmetric-eigenvalue material in Section 2.1 for Part 2, and Section 2.3 for Part 3. After these lessons, the singular value decomposition and conditioning sections explain more systematically the geometry and error amplification already seen in the examples.